General instructions for all candidates

- Complete the core requirements first. Extra features are optional.
- Preferred language: Python
- Use Pandas, NumPy, scikit-learn, and other relevant libraries.
- Submission requirements
- Source code, README with execution instructions, sample output, and results from the
- provided test cases.
- Important: Candidates may use synthetic data if no dataset is supplied. For tasks involving machine learning, the expected predictions are based on the trained model and test data; they do not need to match one universal numerical answer. The test cases below specify the behaviour that must be demonstrated.

Problem: Product Review Sentiment Analysis

NLP · Sentiment Classification

Problem statement

Create an application that determines whether a product review expresses a POSITIVE or NEGATIVE sentiment. The system must use a labelled review dataset to train a model and must support predictions on new reviews.

Requirements

1. Load a dataset with review_text and sentiment columns.

2. Normalize text and handle empty reviews.

3. Train a text classifier using TF-IDF and Logistic Regression, Naive Bayes, or a similar method.

4. Evaluate performance using accuracy, precision, recall, and F1-score.

5. Accept a new review and return its predicted sentiment.

6. Demonstrate predictions on reviews not used for training.

Test cases

Test case	Input	Expected behaviour
TC1	“The product is excellent and works perfectly”	POSITIVE, provided the model has learned similar examples
TC2	“The product broke after one day”	NEGATIVE, provided the model has learned similar examples
TC3	Empty review	Reject or return a clear validation message
TC4	Review containing emojis and punctuation	Process without crashing
TC5	A completely new review	Return a prediction
TC6	Review containing both positive and negative statements	Return the model's predicted sentiment without claiming perfect understanding of context

Acceptance criteria: A working sentiment classifier, evaluation metrics, and a function for predicting sentiment on unseen reviews.